

Supplementary File S3

RQ3: Application Scope, Subsystem Coverage, and Under-Reviewed Areas

Supporting material for *What the Survey Literature Says About Reinforcement Learning for Building Control: A Critical Synthesis*

Scope and coding scale

This supplementary file provides the survey-level evidence audit for RQ3. It reports application-scope scores, subsystem and application coverage, gap-signal scores, coverage imbalances, and concise survey-level answers. RQ3 applies the same shared evidence-intensity scale: the scope matrix codes how visibly each application scope structures the survey, the subsystem matrix codes how visibly each subsystem is covered, and the gap-signal matrix codes how visibly each subsystem is framed as under-reviewed or requiring future work.

Score	Evidence-intensity interpretation
0	Not addressed
1	Mentioned
2	Discussed
3	Structuring role

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S3.1 Reinforcement learning for building controls: The opportunities and challenges [1]

Table S3.1: RQ3 recoded audit table for Wang & Hong (2020). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	HVAC scope	Occupant scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Wang and Hong (2020)	3	2	0	3	2	The survey has a broad building-control scope and reviews RL applications across multiple building-control subjects. Broad scope receives score 3 because the review is explicitly framed around RL for building controls rather than a single subsystem. HVAC scope receives score 2 because HVAC is the most visible subject of control in the reviewed literature, but the survey itself is not limited to HVAC. Occupant-behavior scope receives score 0 because occupant behavior is not the declared scope of the paper. Multi-subsystem scope receives score 3 because the paper covers HVAC, batteries, home-appliance scheduling, domestic hot water, thermal energy storage, combined heat and power, windows, lighting, and related building-control domains. Deployment scope receives score 2 because real-building implementation is discussed, including the limited share of RL controllers implemented in real buildings, but deployment is not the sole organizing focus of the review.

Table S3.2: RQ3 survey-scope scores for Wang & Hong (2020), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	HVAC-focused scope	Occupant-behavior scope	Multi-subsystem scope	Deployment/real-building scope
Wang & Hong (2020)	3	2	0	3	2

Table S3.3: RQ3 subsystem/application coverage scores for Wang & Hong (2020), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Windows
Wang & Hong (2020)	3	2	1	2	0	2	1	2	1	1	1	1

Table S3.4: RQ3 gap-signal scores for Wang & Hong (2020), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Windows
Wang & Hong (2020)	0	0	2	0	1	0	1	1	3	3	3	0

Table S3.5: RQ3 summary table for Wang & Hong (2020), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Wang & Hong (2020)	Broad and HVAC-heavy	HVAC is the most visible subsystem, while batteries and home-appliance scheduling appear as secondary recurring domains. Non-thermal and larger-scale coordination areas are less consistently developed.	The paper supports a conservative RQ3 reading in which HVAC dominance is a recurring survey-level pattern, while visual comfort, IAQ, occupant-feedback-based control, multi-agent community/campus/urban applications, and grid-operation applications remain comparatively under-reviewed or are mainly framed as future needs.

Table S3.6: One-line RQ3 answer for Wang & Hong (2020), after recoding.

Survey	RQ3 answer
Wang and Hong (2020)	Wang & Hong (2020) show that RL-for-building-controls survey coverage is broad but HVAC-heavy. Batteries and appliance scheduling appear as secondary recurring domains, while visual comfort, IAQ, occupant-feedback-based control, and multi-agent community-, campus-, urban-, and grid-oriented applications are less consistently covered and are often framed as future needs.

S3.2 The reinforcement learning method for occupant behavior in building control: A review [2]

Table S3.7: RQ3 recoded audit table for Han et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	HVAC scope	Occupant scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Han et al. (2021)	0	2	3	1	0	The survey is specialized around occupant behavior in building control rather than broad building-control coverage. Broad scope receives score 0 because the review does not aim to cover all RL-for-building subsystems. HVAC scope receives score 2 because HVAC dominates the occupant-behavior RL studies reviewed, although the paper is not declared as HVAC-only. Occupant-behavior scope receives score 3 because occupant behavior is the declared and structuring scope of the review. Multi-subsystem scope receives score 1 because lighting, windows, vessel or water-heating-related control, and one photovoltaic-related case appear, but these remain minor compared with HVAC. Deployment scope receives score 0 because real-building deployment is not a central scope dimension of the review.

Table S3.8: RQ3 survey-scope scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	HVAC-focused scope	Occupant-behavior scope	Multi-subsystem scope	Deployment/real-building scope
Han et al. (2021)	0	2	3	1	0

Table S3.9: RQ3 subsystem/application coverage scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	PV/renewables	Windows
Han et al. (2021)	3	1	1	1	1

Table S3.10: RQ3 gap-signal scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	PV/renewables	Windows
Han et al. (2021)	0	3	3	0	3

Table S3.11: RQ3 summary table for Han et al. (2021), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Han et al. (2021)	Specialized and HVAC-heavy	Within the occupant-behavior RL subliterature, HVAC is the dominant subsystem. Lighting, window, and vessel or water-heating-related controls appear, but are explicitly characterized as relatively uncommon.	The paper supports an RQ3 reading in which occupant-behavior-aware RL remains concentrated on HVAC, while non-HVAC applications are secondary and are framed more as future research directions than as mature, well-covered review areas.

Table S3.12: One-line RQ3 answer for Han et al. (2021), after recoding.

Survey	RQ3 answer
Han et al. (2021)	Han et al. (2021) show a clear HVAC-dominant pattern within RL studies on occupant behavior in building control. Lighting, windows, and vessel or water-heating-related applications are present but minor, and the paper explicitly frames these non-HVAC directions as relatively uncommon areas requiring further study.

S3.3 Systematic review on deep reinforcement learning-based energy management for different building types [3]

Table S3.13: RQ3 recoded audit table for Shaqour & Hagishima (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	HVAC scope	Building-type scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Shaqour and Hagishima (2022)	3	2	3	3	2	The survey has a broad DRL-based BEMS scope and is structured around multiple building types. Broad scope receives score 3 because the paper systematically reviews DRL-based energy management across building applications rather than a single subsystem. HVAC scope receives score 2 because HVAC is highly recurring across building types, but the paper is not limited to HVAC. Building-type scope receives score 3 because residential, office, educational, data-centre, and other building categories structure the review. Multi-subsystem scope receives score 3 because the paper compares controlled appliances and systems, renewable-energy integration, demand response, PV, ESS, EVs, appliances, TES, blinds, lighting, and related building-control elements across building types. Deployment scope receives score 2 because real-world implementation and validation are discussed, including the limited share of real implementations, but deployment is not the sole organizing scope of the review.

Table S3.14: RQ3 survey-scope scores for Shaqour & Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	HVAC-focused scope	Building-type/MORL scope	Multi-subsystem scope	Deployment/real-building scope
Shaqour & Hagishima (2022)	3	2	3	3	2

Table S3.15: RQ3 subsystem/application coverage scores for Shaqour & Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Other niche/Windows
Shaqour & Hagishima (2022)	3	1	1	2	1	2	2	3	2	1	2	1

Table S3.16: RQ3 gap-signal scores for Shaqour & Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Other niche/Windows
Shaqour & Hagishima (2022)	0	1	1	0	2	2	2	0	2	1	3	1

Table S3.17: RQ3 summary table for Shaqour & Hagishima (2022), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Shaqour & Hagishima (2022)	Broad, building-type-centered, and HVAC-heavy	HVAC and cooling are the most consistently visible control targets across building types. Residential studies show the broadest non-HVAC spread through demand response, ESS, PV, appliances, and some EV integration, while office, educational, and especially data-centre studies are more thermally centered.	The paper supports an RQ3 reading in which HVAC remains the most recurrent subsystem across DRL-BEMS surveys, but subsystem diversity depends strongly on building type. Residential studies diversify into DR, PV, ESS, and appliances, whereas EVs, renewable/storage/DR integration in data centres, and multi-building or district-scale coordination remain weaker or explicitly less explored.

Table S3.18: One-line RQ3 answer for Shaqour & Hagishima (2022), after recoding.

Survey	RQ3 answer
Shaqour and Hagishima (2022)	Shaqour & Hagishima (2022) show that DRL-BEMS coverage is broad but still HVAC-heavy across building types. Residential studies have the widest subsystem spread, including demand response, PV, ESS, appliances, and some EV integration, while office, educational, and data-centre studies remain more thermally centered and multi-building or district-scale coordination is explicitly less explored.

S3.4 Reinforcement learning for HVAC control in intelligent buildings: A technical and conceptual review [4]

Table S3.19: RQ3 recoded audit table for Al Sayed et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	HVAC scope	Building-type/MORL scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Al Sayed et al. (2024)	0	3	0	1	2	The survey is explicitly specialized around reinforcement learning for HVAC control rather than broad subsystem comparison. Broad scope receives score 0 because the review does not aim to cover all RL-for-building subsystems. HVAC scope receives score 3 because HVAC is the declared and structuring scope of the review. Building-type/MORL scope receives score 0 because building type and MORL are not the paper’s main organizing axes. Multi-subsystem scope receives score 1 because air quality, CO ₂ , domestic hot water, demand response, PV, and battery-related elements appear only as auxiliary HVAC-linked cases rather than coequal subsystem domains. Deployment scope receives score 2 because real-building implementation, simulation-to-reality mismatch, and generalization are discussed, but deployment is not the declared scope of the survey.

Table S3.20: RQ3 survey-scope scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	HVAC-focused scope	Building-type/MORL scope	Multi-subsystem scope	Deployment/real-building scope
Al Sayed et al. (2024)	0	3	0	1	2

Table S3.21: RQ3 subsystem/application coverage scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Other niche/Windows
Al Sayed et al. (2024)	3	1	0	0	0	1	1	0	1	1	0	0

Table S3.22: RQ3 gap-signal scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Other niche/Windows
Al Sayed et al. (2024)	0	1	0	0	0	1	1	0	1	1	0	0

Table S3.23: RQ3 summary table for Al Sayed et al. (2024), after recoding.			
Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Al Sayed et al. (2024)	Specialized and HVAC-centered	The review is HVAC-dominant by design. Non-HVAC elements such as air quality, CO ₂ , domestic hot water, demand response, PV, and battery-related cases appear mainly as auxiliary HVAC-linked formulation elements rather than as standalone subsystem domains.	For RQ3, this paper should be interpreted as evidence of scope-driven HVAC concentration rather than as a broad subsystem survey. Its strongest explicit gaps concern real deployment and generalization, while subsystem-level omissions beyond HVAC are mostly outside the declared scope.

Table S3.24: One-line RQ3 answer for Al Sayed et al. (2024), after recoding.	
Survey	RQ3 answer
Al Sayed et al. (2024)	Al Sayed et al. (2024) is a specialized RL-for-HVAC review. It shows HVAC as a structuring scope, while IAQ, CO ₂ , domestic hot water, demand response, PV, and battery-related cases appear only as secondary HVAC-linked elements. Its main gaps concern real-building deployment and generalization rather than broad subsystem omission.

S3.5 Multi objectives reinforcement learning for smart buildings: A systematic review of algorithms, applications and future perspectives [5]

Table S3.25: RQ3 recoded audit table for Huynh et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	HVAC scope	MORL scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Huynh et al. (2025)	3	2	3	3	2	The survey has a broad smart-building MORL scope. Broad scope receives score 3 because the paper systematically reviews MORL applications across smart-building domains. HVAC scope receives score 2 because HVAC is highly visible and recurrent, but it is not the sole declared scope. MORL scope receives score 3 because multi-objective reinforcement learning is the declared and structuring focus of the survey. Multi-subsystem scope receives score 3 because the paper’s application taxonomy includes HVAC, lighting, appliance scheduling, EMS, and other applications. Deployment scope receives score 2 because evaluation environments, virtual versus real-world validation, and field implementation are discussed, but deployment is not the sole organizing scope.

Table S3.26: RQ3 survey-scope scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	HVAC-focused scope	Building-type/MORL scope	Multi-subsystem scope	Deployment/real-building scope
Huynh et al. (2025)	3	2	3	3	2

Table S3.27: RQ3 subsystem/application coverage scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Other niche/Windows
Huynh et al. (2025)	3	0	1	2	0	1	1	2	1	1	0	1

Table S3.28: RQ3 gap-signal scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Other niche/Windows
Huynh et al. (2025)	0	2	2	0	2	0	0	0	0	2	0	3

Table S3.29: RQ3 summary table for Huynh et al. (2025), after recoding.			
Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Huynh et al. (2025)	Broad MORL scope with HVAC concentration	The review is broad across smart-building MORL applications, but its most visible application cluster remains HVAC and thermal-comfort-oriented control. Appliance scheduling and EMS form the next substantial tier, while lighting is lighter and many niche smart-building applications remain peripheral.	The paper supports an RQ3 reading in which MORL research remains centered on core energy–comfort control problems, especially HVAC, while broader application domains such as water-related objectives, visual/acoustic comfort, EV-specific concerns, and non-energy building-management functions are weakly represented or framed mainly as future opportunities.

Table S3.30: One-line RQ3 answer for Huynh et al. (2025), after recoding.	
Survey	RQ3 answer
Huynh et al. (2025)	Huynh et al. (2025) portray MORL research in smart buildings as broad but still centered on HVAC and thermal-comfort-oriented control. EMS and appliance scheduling form the next substantive application areas, lighting remains lighter, and many other smart-building applications are peripheral or mainly proposed as future opportunities.

S3.6 Building energy management with reinforcement learning and model predictive control: A survey [6]

Table S3.31: RQ3 recoded audit table for Zhang et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Zhang et al. (2022)	3	3	2	3	2	The survey has a broad building energy management scope and is structured around reinforcement learning, model predictive control, and hybrid BEMS control strategies. Broad scope receives score 3 because the paper frames BEMS as involving HVAC, lighting, household or commercial appliances, EVs, battery energy storage, renewable sources, and the power grid. Main specialized scope receives score 3 because BEMS control with RL, MPC, and RL–MPC combinations is the organizing focus of the survey. Secondary focus scope receives score 2 because HVAC and thermal control are recurring application strands, but the paper is not HVAC-only. Multi-subsystem scope receives score 3 because the paper discusses home energy management, appliance scheduling, domestic hot water, HVAC control, ESS and EV scheduling, commercial-building subsystem control, and multi-agent or community-scale energy management. Deployment scope receives score 2 because real-world exploration cost, simulator and model dependence, historical building data, and practical implementation difficulty are discussed, but deployment is not the sole organizing scope.

Table S3.32: RQ3 survey-scope scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Zhang et al. (2022)	3	3	2	3	2

Table S3.33: RQ3 subsystem/application coverage scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Other RES	Non-battery storage	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Windows
Zhang et al. (2022)	2	1	1	2	2	2	1	0	0	3	2	1	2	0

Table S3.34: RQ3 gap-signal scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Other RES	Non-battery storage	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Windows
Zhang et al. (2022)	0	1	1	0	0	0	1	0	0	0	0	1	1	0

Table S3.35: RQ3 summary table for Zhang et al. (2022), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Zhang et al. (2022)	Broad BEMS and method-centered	The survey covers a broad BEMS application space, with whole-home/building energy management, HVAC, appliances, EVs, ESS, grid/DR, and community-scale settings all visible. HVAC and integrated BEMS are the most recurring strands, but the paper does not make a strong subsystem-imbalance claim.	For RQ3, this paper supports a cautious reading: RL and MPC research in BEMS is broader than HVAC alone, but still strongly organized around thermal control and integrated energy-management problems. Its clearest gaps are methodological—data efficiency, safety, generalization, simulator/model dependence, and learning from historical data—rather than subsystem-specific omissions.

Table S3.36: One-line RQ3 answer for Zhang et al. (2022), after recoding.

Survey	RQ3 answer
Zhang et al. (2022)	Zhang et al. (2022) presents a broad RL–MPC BEMS survey in which integrated home/building energy management and HVAC are the most visible application strands, alongside meaningful coverage of appliances, domestic hot water, EVs, ESS, renewables, grid/DR, and community-scale settings. Its clearest limitations are methodological rather than strong subsystem omissions.

S3.7 Applications of reinforcement learning for building energy efficiency control: A review [7]

Table S3.37: RQ3 recoded audit table for Fu et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Fu et al. (2022)	3	3	2	2	2	The survey has a broad intelligent-building energy-efficiency scope and reviews RL applications across multiple building energy-consumption control objects. Broad scope receives score 3 because the paper frames building energy use through HVAC systems, water heating equipment, lifts, lighting systems, smart-home decision making, photovoltaic and smart-grid control, and other building control applications. Main specialized scope receives score 3 because RL for building energy-efficiency control is the declared organizing focus of the review. Secondary focus scope receives score 2 because HVAC is repeatedly presented as the main energy-consumption target, but the paper is not HVAC-only. Multi-subsystem scope receives score 2 because non-HVAC applications are discussed, including domestic water heaters, dynamic pricing and demand response, PV and smart-grid control, window-related control, smart-home decision making, battery/solar applications, and occupant-related control, but these are not organized as a full subsystem taxonomy. Deployment scope receives score 2 because real-building cost, training burden, unsafe exploration, and simulation-to-reality limitations are discussed, but deployment is not the declared organizing scope.

Table S3.38: RQ3 survey-scope scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Fu et al. (2022)	3	3	2	2	2

Table S3.39: RQ3 subsystem/application coverage scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Other RES	Non-battery storage	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Windows
Fu et al. (2022)	3	2	1	1	0	1	1	0	0	2	2	1	2	1

Table S3.40: RQ3 gap-signal scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Other RES	Non-battery storage	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	Windows
Fu et al. (2022)	0	1	1	1	0	1	1	0	0	0	1	2	3	1

Table S3.41: RQ3 summary table for Fu et al. (2022), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Fu et al. (2022)	Broad but HVAC-centered	HVAC is repeatedly presented as the main energy-consumption and RL-control target, while water heating, lighting/window-related control, smart-home energy management, PV, smart-grid/DR, battery/solar, and occupant-related applications appear as secondary or scattered domains.	The paper supports an RQ3 reading in which HVAC remains the dominant reviewed subsystem within building energy-efficiency control, but non-HVAC applications are present. Its clearest gap signals concern multi-agent control for complex multi-object or larger-scale systems and occupant-related control, rather than sharply defined omission of one subsystem.

Table S3.42: One-line RQ3 answer for Fu et al. (2022), after recoding.

Survey	RQ3 answer
Fu et al. (2022)	Fu et al. (2022) reviews a broad range of RL applications for building energy-efficiency control, but HVAC remains the most visible subsystem. Water heating, lighting/window-related control, smart-home and grid-oriented management, PV, battery/solar, and occupant-related applications appear as secondary domains, while larger-scale multi-agent control is framed as a stronger future need.

S3.8 Reinforcement learning for optimizing renewable energy utilization in buildings: A review on applications and innovations [8]

Table S3.43: RQ3 recoded audit table for Michailidis et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Michailidis et al. (2025)	2	3	3	3	2	The survey is specialized around RL for RES-integrated BEMS, while still covering multiple building-energy subsystems within that scope. Broad scope receives score 2 because the paper is broad within RES-integrated BEMS but is not a generic all-subsystem RL-for-buildings survey. Main specialized scope receives score 3 because RES-integrated BEMS is the declared and structuring scope. Secondary focus scope receives score 3 because PV/solar integration is the dominant RES pattern and structures much of the synthesis. Multi-subsystem scope receives score 3 because the review includes PV, wind turbines, solar water heaters, ground heat pumps, biomass systems, HVAC, domestic hot water, lighting, ESS, EVs, whole-building BEMS, and grid-related applications. Deployment scope receives score 2 because limited real-life experiments and simulation-heavy validation are discussed, but deployment is not the sole organizing scope.

Table S3.44: RQ3 survey-scope scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Michailidis et al. (2025)	2	3	3	3	2

Table S3.45: RQ3 subsystem/application coverage scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	EV/V2H/V2G	Battery/ESS	PV/renewables	Other RES	Non-battery storage	Whole-home/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Michailidis et al. (2025)	2	1	1	1	2	3	1	1	3	2	1

Table S3.46: RQ3 gap-signal scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	EV/V2H/V2G	Battery/ESS	PV/renewables	Other RES	Non-battery storage	Whole-home/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster
Michailidis et al. (2025)	0	1	1	1	0	0	2	3	0	1	3

Table S3.47: RQ3 summary table for Michailidis et al. (2025), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Michailidis et al. (2025)	RES/BEMS-specialized with PV concentration	The review is centered on RES-integrated BEMS, with PV/solar and whole-building BEMS as structuring domains and battery storage as a major recurring component. HVAC appears as a substantial coupled subsystem, while other RES technologies and non-battery storage forms are less visible.	The paper supports an RQ3 reading in which RL research on renewable-energy utilization in buildings is concentrated on PV-driven, battery-aware, whole-building energy management, with HVAC as a recurring companion subsystem. Other RES technologies, non-battery storage forms, and larger-scale district or microgrid coordination remain weaker or explicitly framed as future priorities.

Table S3.48: One-line RQ3 answer for Michailidis et al. (2025), after recoding.

Survey	RQ3 answer
Michailidis et al. (2025)	Michailidis et al. (2025) portray RL research on renewable-energy utilization in buildings as concentrated on RES-integrated BEMS, especially PV/solar and battery-coupled configurations. HVAC is a substantial coupled subsystem, while other RES technologies, non-battery storage, and larger-scale district or microgrid coordination remain less developed or are explicitly identified as future needs.

S3.9 A review of deep reinforcement learning for smart building energy management [9]

Table S3.49: RQ3 recoded audit table for Yu et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Yu et al. (2021)	3	3	2	3	2	The survey has a broad smart-building energy-management scope and is explicitly organized around DRL for SBEM. Broad scope receives score 3 because the paper reviews smart building energy-management applications rather than a single subsystem. Main specialized scope receives score 3 because DRL-based SBEM is the declared and structuring focus of the paper, organized around single building energy subsystems, multiple energy subsystems, and building microgrids. Secondary focus scope receives score 2 because HVAC receives strong attention at the single-subsystem level, but the survey is not HVAC-only. Multi-subsystem scope receives score 3 because the paper covers PV, wind turbines, distributed generation, electric and thermal energy storage, HVAC, lighting, blind and window systems, electric water heaters, EVs, appliances, and building microgrids. Deployment scope receives score 2 because simulation-heavy validation and practical implementation limitations are discussed, but deployment is not the sole organizing scope.

Table S3.50: RQ3 survey-scope scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Yu et al. (2021)	3	3	2	3	2

Table S3.51: RQ3 subsystem/application coverage scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	HVAC	Temperature/thermal comfort	Humidity	IAQ/ventilation	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster	Windows	Thermal storage	Building clusters/local energy communities	Multi-agent BES control
Yu et al. (2021)	3	0	0	1	2	1	2	2	2	2	3	2	2	1	0	0	0

Table S3.52: RQ3 gap-signal scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	HVAC	Temperature/thermal comfort	Humidity	IAQ/ventilation	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster	Windows	Thermal storage	Building clusters/local energy communities	Multi-agent BES control
Yu et al. (2021)	0	0	0	1	0	1	0	0	0	0	0	0	2	1	0	0	0

Table S3.53: RQ3 summary table for Yu et al. (2021), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Yu et al. (2021)	Broad SBEM and scale-based	The review is organized by system scale rather than by a single subsystem. HVAC is the clearest recurring strand at the single-subsystem level, while integrated SBEM and microgrid-oriented settings broaden coverage toward PV, ESS, EVs, appliances, lighting, electric water heaters, and grid interaction.	For RQ3, the paper supports a balanced reading: DRL-based SBEM research is HVAC-heavy at the single-subsystem level, but broader at whole-building and microgrid scales. The strongest gaps are more methodological and scalability-oriented than sharply subsystem-specific.

Table S3.54: One-line RQ3 answer for Yu et al. (2021), after recoding.

Survey	RQ3 answer
Yu et al. (2021)	Yu et al. (2021) review DRL for SBEM through a scale-based lens and show that HVAC dominates the single-subsystem literature, while broader SBEM work substantially covers integrated PV, ESS, EV, appliance, lighting, hot-water, and microgrid configurations. The survey is therefore HVAC-heavy but clearly broader than a dedicated HVAC-only review.

S3.10 A review of reinforcement learning applications to control of heating, ventilation and air conditioning systems [10]

Table S3.55: RQ3 recoded audit table for Sierla et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Sierla et al. (2022)	0	3	3	1	2	The survey is specialized around RL applications to HVAC control rather than broad subsystem coverage. Broad scope receives score 0 because the paper does not aim to review all RL-for-building subsystems. Main specialized scope receives score 3 because HVAC control is the declared and structuring scope. Secondary focus scope receives score 3 because the paper categorizes the literature by zone applicability, action formulation, and indoor-environment variable being controlled, including temperature, humidity, and air quality. Multi-subsystem scope receives score 1 because the paper is not multi-subsystem in the usual building-device sense; its breadth is mainly within HVAC variables and zone structures. Deployment scope receives score 2 because deployability, action abstraction, real HVAC interfacing, and scalability are discussed, but deployment is not the sole organizing scope.

Table S3.56: RQ3 survey-scope scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Sierla et al. (2022)	0	3	3	1	2

Table S3.57: RQ3 subsystem/application coverage scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	HVAC	Temperature/thermal comfort	Humidity	IAQ/ventilation	Single-zone HVAC	Independent-zone HVAC	Interdependent multi-zone HVAC
Sierla et al. (2022)	3	3	1	1	2	1	2

Table S3.58: RQ3 gap-signal scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	HVAC	Temperature/thermal comfort	Humidity	IAQ/ventilation	Single-zone HVAC	Independent-zone HVAC	Interdependent multi-zone HVAC
Sierla et al. (2022)	0	0	1	3	0	1	2

Table S3.59: RQ3 summary table for Sierla et al. (2022), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Sierla et al. (2022)	Specialized HVAC with internal imbalance	The review shows a clear concentration on temperature-based thermal-comfort control within RL-for-HVAC research. Humidity and especially air-quality-oriented formulations are less visible, while interdependent multi-zone control remains challenging because of scalability and realistic HVAC interfacing.	For RQ3, the paper should be interpreted as evidence that HVAC-oriented RL surveys can still be internally imbalanced: temperature-based formulations dominate, while broader indoor-environment variables such as IAQ and humidity receive weaker attention. Its omissions are therefore within HVAC scope, not across unrelated building subsystems.

Table S3.60: One-line RQ3 answer for Sierla et al. (2022), after recoding.

Survey	RQ3 answer
Sierla et al. (2022)	Sierla et al. (2022) show that within RL-for-HVAC research, temperature-based thermal-comfort control dominates the reviewed literature, while humidity and especially air-quality-oriented HVAC control are less developed. The paper’s RQ3 value is therefore mainly in exposing internal imbalance within HVAC-focused RL reviews.

S3.11 Applications of Deep Reinforcement Learning for Home Energy Management Systems: A Review [11]

Table S3.61: RQ3 recoded audit table for Latoń et al. (2024). Scores use the shared 0–3 evidence-intensity scale.						
Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Latoń et al. (2024)	0	3	3	3	2	The survey is specialized around RL and DRL for Home Energy Management Systems rather than broad all-building subsystem coverage. Broad scope receives score 0 because the paper is HEMS-specific. Main specialized scope receives score 3 because HEMS is the declared and structuring scope. Secondary focus scope receives score 3 because demand response, load scheduling, renewable-energy integration, IoT-enabled automation, and residential/prosumer microgrids structure the paper’s application framing. Multi-subsystem scope receives score 3 because the paper discusses HVAC systems, lighting, appliances, electric vehicles, solar panels, battery storage, and smart-grid interaction as HEMS-managed components. Deployment scope receives score 2 because scalability, privacy, data security, heterogeneous system integration, and the need for real applications and case studies are discussed, but deployment is not the sole organizing scope.

Table S3.62: RQ3 survey-scope scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.					
Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Latoń et al. (2024)	0	3	3	3	2

Table S3.63: RQ3 subsystem/application coverage scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.								
Survey	HVAC	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid
Latoń et al. (2024)	1	1	2	1	2	2	3	2

Table S3.64: RQ3 gap-signal scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	HVAC	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid
Latoń et al. (2024)	1	1	0	1	0	0	0	0

Table S3.65: RQ3 summary table for Latoń et al. (2024), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Latoń et al. (2024)	Specialized HEMS orchestration	The review is centered on whole-home energy-management functions, especially demand response, load scheduling, renewable integration, energy storage, smart-grid interaction, and residential/prosumer microgrids. Individual subsystems such as HVAC and lighting appear mainly as components inside broader HEMS coordination.	For RQ3, the paper shows that not all recent RL-for-buildings surveys are HVAC-heavy. Some are instead centered on HEMS-level orchestration, where the main application bias shifts toward scheduling, DSM/DSR, RES-storage coordination, and smart-grid integration. The main gaps are deployment and infrastructure related rather than subsystem-specific.

Table S3.66: One-line RQ3 answer for Latoń et al. (2024), after recoding.

Survey	RQ3 answer
Latoń et al. (2024)	Latoń et al. (2024) portray RL/DRL research in HEMS as centered on whole-home orchestration, especially demand response, load scheduling, renewable integration, storage management, and smart-grid interaction. HVAC, lighting, and other individual subsystems appear mainly as components within that broader HEMS frame rather than as standalone review targets.

S3.12 A review of reinforcement learning for controlling building energy systems from a computer science perspective [12]

Table S3.67: RQ3 recoded audit table for Weinberg et al. (2023). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Weinberg et al. (2023)	0	3	3	1	2	The survey is specialized around RL for Building Energy Systems from a computer-science perspective, with BES narrowed mainly toward thermal systems. Broad scope receives score 0 because this is not a broad all-subsystem review. Main specialized scope receives score 3 because thermal BES and HVAC-related control define the paper’s application scope. Secondary focus scope receives score 3 because the survey is explicitly structured around RL in single buildings, RL in building clusters, and multi-agent aspects. Multi-subsystem scope receives score 1 because the breadth is mainly scale-based rather than device-category-based; diverse non-thermal subsystems such as lighting, appliances, and EVs are not a major part of the review. Deployment scope receives score 2 because transfer learning, sim-to-real mismatch, differing state–action spaces across buildings, data limitations, and practical implementation challenges are discussed, but deployment is not the only organizing scope.

Table S3.68: RQ3 survey-scope scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Weinberg et al. (2023)	0	3	3	1	2

Table S3.69: RQ3 subsystem/application coverage scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	HVAC	Thermal storage	DHW/water	Whole-building thermal BES	Building clusters/local energy communities	Multi-agent BES control
Weinberg et al. (2023)	3	2	1	2	2	3

Table S3.70: RQ3 gap-signal scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.						
Survey	HVAC	Thermal storage	DHW/water	Whole-building thermal BES	Building clusters/local energy communities	Multi-agent BES control
Weinberg et al. (2023)	0	0	1	0	3	3

Table S3.71: RQ3 summary table for Weinberg et al. (2023), after recoding.			
Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Weinberg et al. (2023)	Specialized thermal-BES and scale-based	The review is concentrated on thermal BES and HVAC-related control, especially heating, cooling, ventilation, and thermal storage. Its main broadening beyond single-building thermal control occurs through building-cluster control and multi-agent coordination rather than through extensive coverage of diverse non-thermal sub-systems.	For RQ3, the paper shows that some RL-for-buildings reviews remain strongly centered on thermal system control, with the main next frontier being larger-scale coordination across buildings rather than balanced coverage of lighting, appliances, EVs, and other non-thermal device categories.

Table S3.72: One-line RQ3 answer for Weinberg et al. (2023), after recoding.	
Survey	RQ3 answer
Weinberg et al. (2023)	Weinberg et al. (2023) present a thermal-BES- and HVAC-centered RL review that extends mainly through building-cluster and multi-agent perspectives. The paper therefore shows stronger survey attention to heating, cooling, ventilation, and thermal storage than to a broad range of non-thermal building subsystems.

S3.13 Reinforcement learning: theory and applications in HEMS [13]

Table S3.73: RQ3 recoded audit table for Al-Ani & Das (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Al-Ani and Das (2022)	0	3	2	3	2	The survey is specialized around reinforcement learning for home energy management systems rather than broad all-building subsystem coverage. Broad scope receives score 0 because the paper is HEMS-specific. Main specialized scope receives score 3 because HEMS is the declared and structuring scope. Secondary focus scope receives score 2 because HVAC, fans, and water heaters form the largest single application class, but the survey is not limited to this class. Multi-subsystem scope receives score 3 because the paper divides the reviewed literature into multiple application classes, including HVAC/fans/water heaters, EVs, energy storage, renewable generation, other loads, demand response, and peer-to-peer trading. Deployment scope receives score 2 because scalability, data security, and the need for more real applications and case studies are discussed, but deployment is not the only organizing scope.

Table S3.74: RQ3 survey-scope scores for Al-Ani & Das (2022), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Al-Ani & Das (2022)	0	3	2	3	2

Table S3.75: RQ3 subsystem/application coverage scores for Al-Ani & Das (2022), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	P2P trading
Al-Ani & Das (2022)	3	3	0	2	2	2	2	3	2	0	0	1

Table S3.76: RQ3 gap-signal scores for Al-Ani & Das (2022), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	P2P trading
Al-Ani & Das (2022)	0	0	0	0	1	1	1	0	0	0	0	2

Table S3.77: RQ3 summary table for Al-Ani & Das (2022), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Al-Ani & Das (2022)	Specialized HEMS with multi-class coverage	The review is centered on HEMS orchestration, but within that frame the largest single application class is HVAC, fans, and water heaters. EV/storage/renewable generation and demand response are also substantial recurring domains, while peer-to-peer trading is a smaller tail of the reviewed literature.	For RQ3, the paper shows that RL-for-HEMS surveys are not purely HVAC-only, but still give the strongest single-class emphasis to thermal-device control. They broaden toward storage, renewables, demand response, and grid-facing coordination, while peer-to-peer trading remains less visibly developed.

Table S3.78: One-line RQ3 answer for Al-Ani & Das (2022), after recoding.

Survey	RQ3 answer
Al-Ani and Das (2022)	Al-Ani & Das (2022) portray RL research in HEMS as centered on whole-home orchestration, with HVAC/fans/water-heater control as the largest single application class, substantial additional coverage of EV/storage/renewable generation and demand response, and comparatively limited attention to peer-to-peer energy trading.

S3.14 A review of reinforcement learning for autonomous building energy management [14]

Table S3.79: RQ3 recoded audit table for Mason & Grijalva (2019). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Mason and Grijalva (2019)	3	3	2	3	2	The survey has a broad autonomous building energy management scope and reviews reinforcement learning applications across several building-energy domains. Broad scope receives score 3 because the paper is framed as a comprehensive review of RL for autonomous building energy management. Main specialized scope receives score 3 because autonomous building energy management is the organizing focus of the review. Secondary focus scope receives score 2 because HVAC is the strongest and most recurrent application strand, but the paper is broader than HVAC-only. Multi-subsystem scope receives score 3 because the paper covers HVAC, water heaters, home management systems, lighting, appliances, PV, batteries, EVs, and smart-grid interaction. Deployment scope receives score 2 because simulation dependence, simulator design, and real-world robustness are discussed, but deployment is not the sole organizing scope.

Table S3.80: RQ3 survey-scope scores for Mason & Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Mason & Grijalva (2019)	3	3	2	3	2

Table S3.81: RQ3 subsystem/application coverage scores for Mason & Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	P2P trading
Mason & Grijalva (2019)	3	2	1	2	1	1	1	2	1	1	1	0

Table S3.82: RQ3 gap-signal scores for Mason & Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Lighting/daylighting	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	IAQ/IEQ	Multi-zone/district/cluster	P2P trading
Mason & Grijalva (2019)	0	0	1	0	1	1	1	0	0	1	3	0

Table S3.83: RQ3 summary table for Mason & Grijalva (2019), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Mason & Grijalva (2019)	Broad early BEM and HVAC-led	The review is clearly HVAC-led, with water heaters as a meaningful secondary domain and broader home management systems extending coverage to appliances, lighting, PV, batteries, EVs, and smart-grid interaction. The paper does not provide a strong quantitative subsystem-bias analysis, but its narrative emphasis is visibly heavier on HVAC than on other domains.	For RQ3, the paper supports a reading in which early RL-for-building-energy-management surveys were dominated by HVAC control, while water heating and multi-device home management formed secondary strands. The clearest future need is scaling toward communities of homes rather than balancing coverage across every subsystem.

Table S3.84: One-line RQ3 answer for Mason & Grijalva (2019), after recoding.

Survey	RQ3 answer
Mason and Grijalva (2019)	Mason & Grijalva (2019) present a broad RL-for-building-energy-management review in which HVAC is the dominant application area, water heaters form a clear secondary domain, and broader home-management studies extend coverage to appliances, lighting, PV, batteries, and EV-related coordination without displacing HVAC as the main emphasis.

S3.15 An overview of reinforcement learning-based approaches for smart home energy management systems with energy storages [15]

Table S3.85: RQ3 recoded audit table for Pinthurat et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Pinthurat et al. (2024)	0	3	3	3	3	The survey is specialized around reinforcement-learning-based smart home energy management systems with energy storage. Broad scope receives score 0 because the paper is not a broad all-building subsystem survey. Main specialized scope receives score 3 because SHEMS with energy storage is the declared and structuring scope. Secondary focus scope receives score 3 because energy-storage integration structures the entire review, including battery type, state of charge, state of health, life cycle, degradation, storage constraints, cost, efficiency, operation mode, and pricing mechanisms. Multi-subsystem scope receives score 3 because the smart-home architecture includes energy storage, renewable energy, plug-in EVs, controllable appliances such as AC and water heaters, uncontrollable appliances, smart meters, and grid interaction. Deployment scope receives score 3 because real-world testing and validation are treated as a dedicated future direction, with field trials emphasized as essential for bridging theory and practice.

Table S3.86: RQ3 survey-scope scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Pinthurat et al. (2024)	0	3	3	3	3

Table S3.87: RQ3 subsystem/application coverage scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster	P2P trading
Pinthurat et al. (2024)	1	1	2	1	3	2	3	2	1	0

Table S3.88: RQ3 gap-signal scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	Appliances	EV/V2H/V2G	Battery/ESS	PV/renewables	Whole-home/HEMS/BEMS	Grid/DR/microgrid	Multi-zone/district/cluster	P2P trading
Pinthurat et al. (2024)	1	1	0	1	0	0	0	3	3	0

Table S3.89: RQ3 summary table for Pinthurat et al. (2024), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Pinthurat et al. (2024)	Specialized SHERMS and storage-centered	The review is centered on energy-storage-integrated smart-home orchestration, with battery/energy storage as the most visible application component and strong linkage to renewable integration, demand-side management, and grid-aware operation. HVAC, AC, and water heaters appear mainly as controllable appliances inside the smart-home architecture rather than as standalone review domains.	For RQ3, the paper shows that some recent RL-for-homes reviews are not HVAC-dominant, but instead storage-centered. The clearest under-reviewed directions are demand-side management integration and multi-agent or multi-home coordination, both framed as future research needs.

Table S3.90: One-line RQ3 answer for Pinthurat et al. (2024), after recoding.

Survey	RQ3 answer
Pinthurat et al. (2024)	Pinthurat et al. (2024) portray RL research in smart homes as centered on energy-storage-integrated SHERMS orchestration, with battery/energy storage as the structuring application focus, strong coupling to renewable integration and grid-aware management, and comparatively limited use of multi-agent or multi-home coordination approaches.

S3.16 Reinforcement learning for control and optimization of real buildings: Identifying and addressing implementation hurdles [16]

Table S3.91: RQ3 recoded audit table for Kannari et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Kannari et al. (2025)	0	3	3	1	3	The survey is specialized around reinforcement-learning-based control and optimization in real buildings rather than broad all-building subsystem coverage. Broad scope receives score 0 because the paper is not a broad subsystem-by-subsystem review. Main specialized scope receives score 3 because real-building RL deployment is the declared and structuring scope of the review. Secondary focus scope receives score 3 because the reviewed real-building deployment literature is strongly organized around HVAC, heating, cooling, ventilation, and related thermal services. Multi-subsystem scope receives score 1 because the paper includes only scattered broader cases, such as domestic hot water, windows plus air cleaning, HVAC plus lighting and plug loads, and one floor-heating case coupled with an EV. Deployment scope receives score 3 because deployment in actual buildings is central to the paper, including real-world implementation conditions, control interval, deployment mode, validation setting, pretraining, safety, and practical adoption concerns.

Table S3.92: RQ3 survey-scope scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Kannari et al. (2025)	0	3	3	1	3

Table S3.93: RQ3 subsystem/application coverage scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	IAQ/ventilation/humidity	Lighting/plug loads	EV/V2H/V2G	Whole-building/BMS holistic
Kannari et al. (2025)	3	1	1	1	1	1

Table S3.94: RQ3 gap-signal scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	HVAC	DHW/water	IAQ/ventilation/humidity	Lighting/plug loads	EV/V2H/V2G	Whole-building/BMS holistic
Kannari et al. (2025)	0	1	3	1	1	3

Table S3.95: RQ3 summary table for Kannari et al. (2025), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Kannari et al. (2025)	Specialized real-building deployment and HVAC-centered	The review of real-building RL deployments is heavily concentrated on HVAC and related thermal-control applications, often in single-zone settings and with comfort represented mainly through indoor temperature. Broader services such as domestic hot water, lighting, plug loads, EV coupling, and richer IAQ-focused or personalized comfort control appear only in scattered cases.	For RQ3, the paper is best interpreted as evidence that practical real-building RL deployments remain strongly HVAC-centric. The strongest under-reviewed directions are richer IEQ/IAQ-aware control and more holistic whole-building BMS perspectives beyond heating and cooling alone.

Table S3.96: One-line RQ3 answer for Kannari et al. (2025), after recoding.

Survey	RQ3 answer
Kannari et al. (2025)	Kannari et al. (2025) show that real-building RL deployments are still dominated by HVAC and thermal-control applications. IAQ-rich control, domestic hot water, lighting and plug loads, EV-coupled cases, and especially holistic whole-building BMS control remain sparse or explicitly identified as weakly represented directions.

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Table S3.97: RQ3 recoded audit table for Chatterjee & Khovalyg (2023). Scores use the shared 0–3 evidence-intensity scale.

Survey	Broad scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment scope	Recoded evidence justification
Chatterjee and Khovalyg (2023)	0	3	3	0	2	The survey is specialized around reinforcement-learning-based HVAC controls for dynamic indoor thermal environments rather than broad all-building subsystem coverage. Broad scope receives score 0 because the paper is not a general subsystem survey. Main specialized scope receives score 3 because HVAC-based thermal-environment control is the declared and structuring scope. Secondary focus scope receives score 3 because dynamic indoor thermal environment control, including temperature step changes, drifts, thermal comfort, adaptation, and the energy–comfort–health framing, structures the paper. Multi-subsystem scope receives score 0 because broader non-HVAC services are outside the review’s scope. Deployment scope receives score 2 because real-building implementation, co-simulation, benchmarking, and open virtual testbed needs are discussed, but deployment is not the sole organizing scope.

Table S3.98: RQ3 survey-scope scores for Chatterjee & Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	Broad building-control scope	Main specialized scope	Secondary focus scope	Multi-subsystem scope	Deployment/real-building scope
Chatterjee & Khovalyg (2023)	0	3	3	0	2

Table S3.99: RQ3 subsystem/application coverage scores for Chatterjee & Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	HVAC	Dynamic temperature control/drifts	Humidity/air speed	IAQ	Broader non-HVAC services
Chatterjee & Khovalyg (2023)	3	3	1	1	0

Table S3.100: RQ3 gap-signal scores for Chatterjee & Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	HVAC	Dynamic temperature control/drifts	Humidity/air speed	IAQ	Broader non-HVAC services
Chatterjee & Khovalyg (2023)	0	3	1	1	0

Table S3.101: RQ3 summary table for Chatterjee & Khovalyg (2023), after recoding.

Survey	Coverage pattern	Strongest imbalance identified	Main RQ3 consequence
Chatterjee & Khovalyg (2023)	Specialized dynamic thermal-environment and HVAC-centered	The review is concentrated on HVAC-based thermal-environment control, especially the possibility of using RL to create dynamic temperature variations instead of static deadband control. Within that specialized scope, temperature is the dominant controlled variable, while humidity, air speed, and indoor air quality are acknowledged but much less central.	For RQ3, the paper is best interpreted as evidence that a specialized strand of RL-for-buildings research is strongly HVAC- and temperature-centric. Even within this strand, RL for genuinely dynamic indoor thermal environments remains underdeveloped compared with conventional energy-saving and thermal-comfort optimization goals.

Table S3.102: One-line RQ3 answer for Chatterjee & Khovalyg (2023), after recoding.

Survey	RQ3 answer
Chatterjee and Khovalyg (2023)	Chatterjee & Khovalyg (2023) show that this review strand is strongly centered on HVAC-driven thermal control, with dynamic temperature control as the structuring topic and temperature as the dominant controlled dimension. RL for genuinely dynamic indoor thermal environments remains comparatively underdeveloped, while broader non-thermal building services are outside the review’s scope.

References

- [1] Zhe Wang and Tianzhen Hong. Reinforcement learning for building controls: The opportunities and challenges. *Applied Energy*, 269:115036, 2020.
- [2] Mengjie Han, Jing Zhao, Xingxing Zhang, Jingchun Shen, and Yu Li. The reinforcement learning method for occupant behavior in building control: A review. *Energy and Built Environment*, 2(2):137–148, 2021.
- [3] Ayas Shaqour and Aya Hagishima. Systematic review on deep reinforcement learning-based energy management for different building types. *Energies*, 15(22):8663, 2022.
- [4] Khalil Al Sayed, Abhinandana Boodi, Roozbeh Sadeghian Broujeny, and Karim Beddiar. Reinforcement learning for hvac control in intelligent buildings: A technical and conceptual review. *Journal of Building Engineering*, 95:110085, 2024.
- [5] Thi Ngoc Yen Huynh, Anh Tuan Nguyen, Yonghan Ahn, Bee Lan Oo, and Benson TH Lim. Multi objectives reinforcement learning for smart buildings: A systematic review of algorithms, applications and future perspectives. *Energy and Buildings*, page 116045, 2025.
- [6] Huiliang Zhang, Sayani Seal, Di Wu, François Bouffard, and Benoit Boulet. Building energy management with reinforcement learning and model predictive control: A survey. *IEEE Access*, 10:27853–27862, 2022.
- [7] Qiming Fu, Zhicong Han, Jianping Chen, You Lu, Hongjie Wu, and Yunzhe Wang. Applications of reinforcement learning for building energy efficiency control: A review. *Journal of Building Engineering*, 50:104165, 2022.
- [8] Panagiotis Michailidis, Iakovos Michailidis, and Elias Kosmatopoulos. Reinforcement learning for optimizing renewable energy utilization in buildings: A review on applications and innovations. *Energies*, 18(7):1724, 2025.
- [9] Liang Yu, Shuqi Qin, Meng Zhang, Chao Shen, Tao Jiang, and Xiaohong Guan. A review of deep reinforcement learning for smart building energy management. *IEEE Internet of Things Journal*, 8(15):12046–12063, 2021.
- [10] Seppo Sierla, Heikki Ihasalo, and Valeriy Vyatkin. A review of reinforcement learning applications to control of heating, ventilation and air conditioning systems. *Energies*, 15(10):3526, 2022.
- [11] Dominik Latoń, Jakub Grela, and Andrzej Ożadowicz. Applications of deep reinforcement learning for home energy management systems: A review. *Energies*, 17(24):6420, 2024.
- [12] David Weinberg, Qian Wang, Thomas Ohlson Timoudas, and Carlo Fischione. A review of reinforcement learning for controlling building energy systems from a computer science perspective. *Sustainable cities and society*, 89:104351, 2023.
- [13] Omar Al-Ani and Sanjoy Das. Reinforcement learning: theory and applications in hems. *Energies*, 15(17):6392, 2022.
- [14] Karl Mason and Santiago Grijalva. A review of reinforcement learning for autonomous building energy management. *Computers & Electrical Engineering*, 78:300–312, 2019.
- [15] Watcharakorn Pinthurat, Tossaporn Surinkaew, and Branislav Hredzak. An overview of reinforcement learning-based approaches for smart home energy management systems with energy storages. *Renewable and Sustainable Energy Reviews*, 202:114648, 2024.
- [16] Lotta Kannari, Nina Wessberg, Sara Hirvonen, Julia Kantorovitch, and Satu Paiho. Reinforcement learning for control and optimization of real buildings: Identifying and addressing implementation hurdles. *Journal of Building Engineering*, page 112283, 2025.
- [17] Arnab Chatterjee and Dolaana Khovalyg. Dynamic indoor thermal environment using reinforcement learning-based controls: Opportunities and challenges. *Building and environment*, 244:110766, 2023.